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MIL-W-28517B(YD)
20 November 1991
SUPERSEDING
MIL-W-28517A(YD)
1 April 1981

MILITARY SPECIFICATION

WATER SUPPLY SYSTEM, AUTOMATIC

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers reciprocating pump water supply systems complete with tank and automatic controls.

1.2 Classification. The water supply system is of the following size, class, style, and type.

Size 5 - 500 gallons per hour (gph), minimum, at cut-in pressure.

Class B - 40 pounds per square inch (psi) cut-in; 60 psi cut-out pressure.

Style I - Reciprocating pump.

Type a - 25 foot suction lift.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320

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2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6 2).

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-F-3541 - Fittings, Lubrication.
MIL-P-16789 - Pumps (Including Prime Movers and Support Items); Packaging of

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Unless otherwise indicated, copies of military specifications and standards are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code.
Section VIII, Division 1 - Rules for Construction of Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

HYDRAULIC INSTITUTE

Hydraulic Institute Standards.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, 14600 Detroit Ave., Cleveland, OH 44107.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Controls and Systems.
NEMA ICS 2 - Industrial Control Devices, Controllers and Assemblies.
NEMA ICS 6 - Enclosures for Industrial Controls and Systems.
NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence , Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The water supply system unit shall include a horizontal, reciprocating pump complete with vertical storage tank, air volume control, electric motor drive, and automatic controls. The motor and pump shall be mounted on a common bedplate. Each pump shall be equipped with all connecting lines and accessories required for proper operation.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term “recovered materials” means materials

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which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.3.1 Selection of materials for pump parts. Unless otherwise specified herein, the selection of materials used in the manufacture of the pump shall be in accordance with the Hydraulic Institute Standards. The manufacturer shall select the materials which shall minimize galvanic corrosion to the greatest extent possible.

3.4 Design and construction. The automatic water supply system shall be size 5, 500 gph minimum at cut-in pressure; class B, 40 psi cut-in, 60 psi cut-out pressure; style I, reciprocating pump; type a, 25 foot maximum suction lift. All lubrication fittings shall conform to MIL-F-3541. The unit shall be designed and constructed to withstand the strains, jars, vibrations, and other conditions associated with shipping, storage, and installation; and shall be complete so that when installed, it can be used for any operation for its intended purpose. The unit and accessories shall be designed and constructed to prevent conditions hazardous to personnel or deleterious to equipment and shall permit easy and ready accessibility for replacement of accessories, maintenance, service, and adjustments.

3.4.1 Pump. The pump furnished with the unit shall be of the power reciprocating type, designed and constructed in accordance with the Hydraulic Institute Standards. The pump shall be self-priming after initial priming.

3.4.1.1 Liquid end assembly. Liquid end assembly shall be either the valve plate or the side pot type configuration The liquid cylinder shall be of close-grained cast iron, lined with bronze or brass renewable liners. Cylinder heads or covers shall be flanged. Valve assembly shall be of the disk or wing type and shall be constructed of rubber, bronze, or corrosion-resistant steel, as specified (see 6.2). The valves shall operate with minimum lost motion or chatter and shall not become permanently distorted when subjected to full pressure load. Valve seats shall be replaceable. Piston shall be of the cup type; brass castings inside and brass plates outside the cup leathers shall hold the leathers in shape. Bearings may be roller or sleeve-type, as applicable to design requirements.

3.4.1.2 lower end. The power end shall be of the horizontal plunger power type. Unless otherwise specified (see 6.2), the reciprocating pump shall be V-belt driven through a pulley functioning as a flywheel.

3.4.1.3 Relief valve. A pressure relief valve sized to discharge the full-rated water flow of the pump at the selected relied pressure shall be installed at the pump discharge or on the tank. A pressure relief valve that carries the appropriate ASME code symbol stamp will be acceptable.

3.5 Performance. The system shall be capable of meeting the capacity and lift requirements without exceeding the continuous horsepower rating of the pump driver.

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3.5.1 Hydrostatic pressure. The casing of the pump shall withstand a hydrostatic pressure equal to 1.5 times the discharge pressure or the tot dynamic head, whichever is greater.

3.6 Controls.

3.6.1 Automatic air volume control. An automatic air-volume control floating diaphragm separator shall be installed to maintain the proper le air above the water in the tank.

3.6.2 Automatic air volume control. An automatic pressure switch shall incorporated in the design of the unit. The switch shall be calibrated a adjusted to start the motor when tank pressure falls below the 40 psi min cut-in pressure, and stop the motor when pressure reaches the 60 psi. maxi cut-out pressure.

3.7 Motor. Unless otherwise specified (see 6.2), electric motor shal designed to operate on 115 volt, single phase, 60-cycle, altercating curr Electric motors shall conform to NEMA MG 1 and shall be rated for continu duty with starting torque and speed rating suitable for the application. motor shall have a horsepower rating adequate for operation of the specif pump within allowable temperature limits. The horsepower rating shall be less than the maximum brake horsepower requirement of the pump under any conditions of operation. Motor starters shall be designed and constructe accordance with NEMA ICS 1 and NEMA ICS 2; and shall have enclosures in accordance with NEMA ICS 6 with NEMA 1, NEMA 3R, or NEMA 12 type, as spec (see 6.2).

3.8 Tank. Unless otherwise specified (see 6.2), the water supply uni include a 120-gallon capacity, hydro-pneumatic vertical tank. Unless oth specified (see 6.2), the tank shall be the supercharged-type, made of ste hot-dip galvanized after fabrication. The tank shall be constructed in accordance with ASME Boiler and Pressure Vessel Code, Section VIII, Divisi Acceptable evidence of the tank being constructed in accordance with ASME and Pressure Vessel Code will be the ASME Code Symbol stamp.

3.9 Bedplate. Each pump and driver shall be bolted to a common bedpl cast iron or fabricated steel design to withstand strains and to maintain and driver in alinement.

3.10 Safety. All mechanical and electrical parts that are of such na so located as to become a hazard to operating or maintenance personnel sha enclosed or properly guarded. The pump shall comply with the OSHA regulat Part 1910.95 for noise and Part 1910.219 for guarding.

3.11 Interchangeability. All units of the same classification furnis with similar options under a specific contract shall be identical to the e necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.12 Identification marking. Identification shall be permanently and legibly marked directly on the water supply system or on a corrosion-resisting metal plate securely attached to the water supply system at the source of manufacture. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.13 Instruction plates. The pump shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.14 Dissimilar metals. Metals dissimilar with respect to the galvanic scale shall not be used unless separated by an insulating material which will avoid electrolytic corrosion.

3.15 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and Spatter, loose mill scale water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.16 Workmanship.

3.16.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.16.2 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and uniform size for the same diameter of rivet, Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member,

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3.16.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.16.5 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale , paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified In the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete automatic water supply system when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5 and when specified (see 6.2), the packaging inspection of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

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4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the test of 4.5.2, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete automatic water supply system. All pumps offered for delivery at one time shall be considered a lot for the purpose of inspection. Guidance for inspection level and Acceptable Quality Level (AQL) is provided in 6.5.

4.4 Examination. Each unit shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests.

4.5.1 Performance test. One pump, complete with motor and tank, shall be tested to meet the capacity and lift requirements of 1.2 and 3.4. The pump shall be operated continuously for two hours. The test shall demonstrate the ability of the pump to handle water at the capacity, suction lift, and discharge pressure requirements specified Failure of the pump to meet any performance requirements shall constitute failure of the test.

4.5.2 Hydrostatic Pressure test. All pump casings produced shall be tested to verify compliance with 3.5.1. All pump casings shall be subjected to a hydrostatic pressure equal to 1.5 times the pump discharge pressure or the total dynamic head, whichever is greater, for a period of not less than five minutes. Minor leakage through the mechanical seal shall not be cause for rejection Any damage or permanent deformation to the pump casing or leakage in the casing or through the casing gasket shall constitute failure of the test.

4.5.3 Discharge Pressure. All tests shall be made with the pump discharging into an air-bound receiver in which constant pressure is maintained equal to the rated discharge pressure of the pump (see 1.2). A test of the pump involving pump discharge through a throttling valve and into an open tank will not be acceptable.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements in section 4 of MIL-P-16789. The inspection shall consist of the quality conformance inspection (see 4.2.2); and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

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5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and m shall be in accordance with the requirements of MIL-P-16789, with the lev presentation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature may be helpful, but is not mandatory.)

6.1 Intended use. Water supply systems covered by this specification intended primarily for use in domestic water supply systems.

6.1.1 Selection factors.

a. Fungus resistant varnish conforming to MIL-V-173 should be use coat electrical components and circuit elements, including ter and circuit connections, when the unit is to be installed in h conditions . Components and elements inherently inert to fungi hermetically sealed enclosures or current-carrying contact sur should not be coated.
b. Electromagnetic interference suppression (EMI), when required, conform to the EMI suppression requirements and test limits as specified in MIL-STD-461.

6.2 Acquisition requirements. Acquisition documents should specify th following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if require specific issue of individual documents referenced (see 2.1.1 ar 2.2).
c. When first article is required for inspection and approval (see and 4.2.1).
d. Material required for valves (see 3,4.1.1).
e. Type of drive for reciprocating pump, if different (see 3.4.1.2
f. Electrical characteristics, if different, and type of motor sta enclosure required (see 3.7).
g. Tank requirements, if different (see 3.8).
h. Color of the finish coat (see 3.15).
i. When a preproduction pack inspection is required and the time f required for submission (see 4.2.1 and 4.6).
j. Level of presentation and level of packing required (see 5.1).

6.3 Data requirements . When this specification is used in an acquisit and data are required to be delivered, the data requirements shall be deve as specified by an approved Data Item Description (DD Form 1664) and deliv in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquis Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) ar invoked and the DD Form 1423 is not used, the data should be delivered by contractor in accordance with the contract or purchase order requirements.

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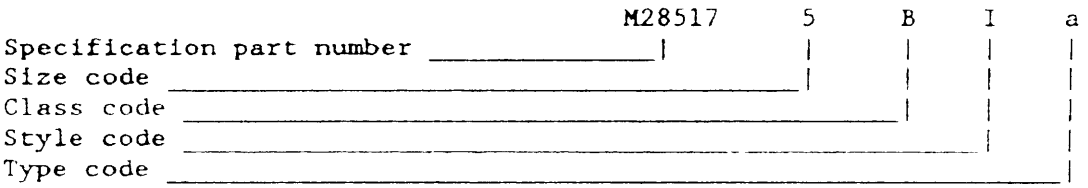
6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Sampling procedures.

6.5.1 Sampling for examination. Recommended inspection level is S-3 and AQL is 2.5 (see 4.3).

6.5.2 Sampling for tests. Recommended inspection level is S-2 and AQL is 2 5 (see 4.3).

6.6 Part or identifying number (PIN). The PIN to be used for the water supply system applied to this specification is created as follows:



6.7 Cross-reference of classification. The classification of water supply systems in this document differs from the previous issue in the following respects:

MIL-W-28517A(YD)	MIL-W-28517B(YD)
Size 3	Deleted.
Size 5	Size 5
Size 7	Deleted.
Class A	Deleted.
Class B	Class B
Style I	Style I
Style II	Deleted.
Style III	Deleted.
Type (a)	Type a
Type (b)	Deleted.
Type (c)	Deleted.
Type (d)	Deleted.
Type (e)	Deleted.
Type (f)	Deleted.
Type (g)	Deleted
Type (h)	Deleted.
Type (l)	Deleted.
Type (n)	Deleted.
Type (p)	Deleted.

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6.8 Subject term (key word) listing.

Reciprocating pump
Water tank

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing Activity:
Navy - YD

(Project 4320-N339)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-W-28517B(YD)	2. DOCUMENT DATE (YYMMDD) 911120

3. DOCUMENT TITLE
WATER SUPPLY SYSTEM, AUTOMATIC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ANNE L. LYONS	b. TELEPHONE (Include Area Code) (1) Commercial: 805-982-5761 (2) AUTOVON: 551-5761
c. ADDRESS (Include Zip Code) COMMANDING OFFICER NAVAL CONSTRUCTION BATTALION CENTER (1564A) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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